

Evaluating NFL Quarterbacks

Team 98: Devesh and Aditya

Link: <https://youtu.be/ReZg--SswjA>

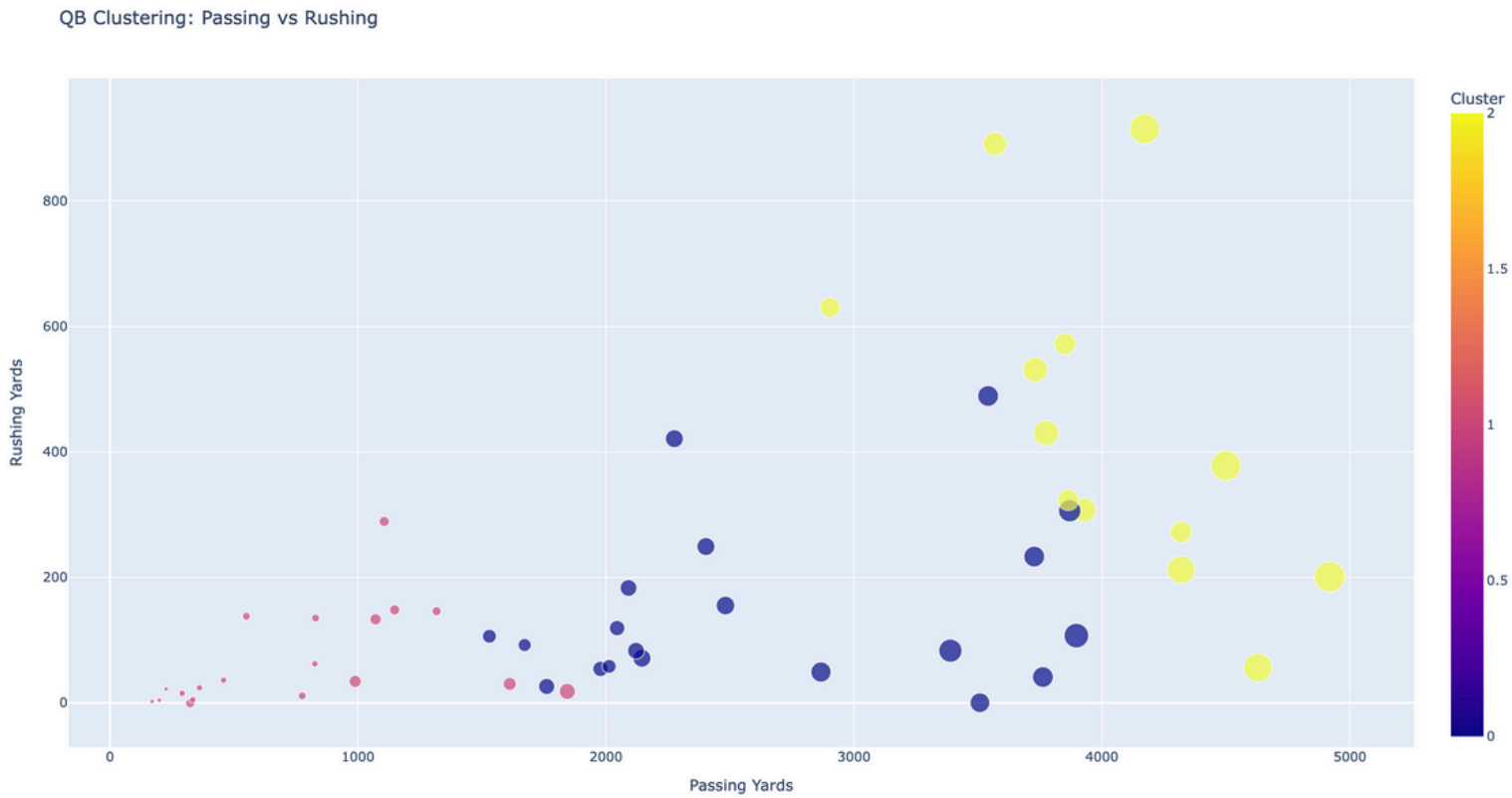
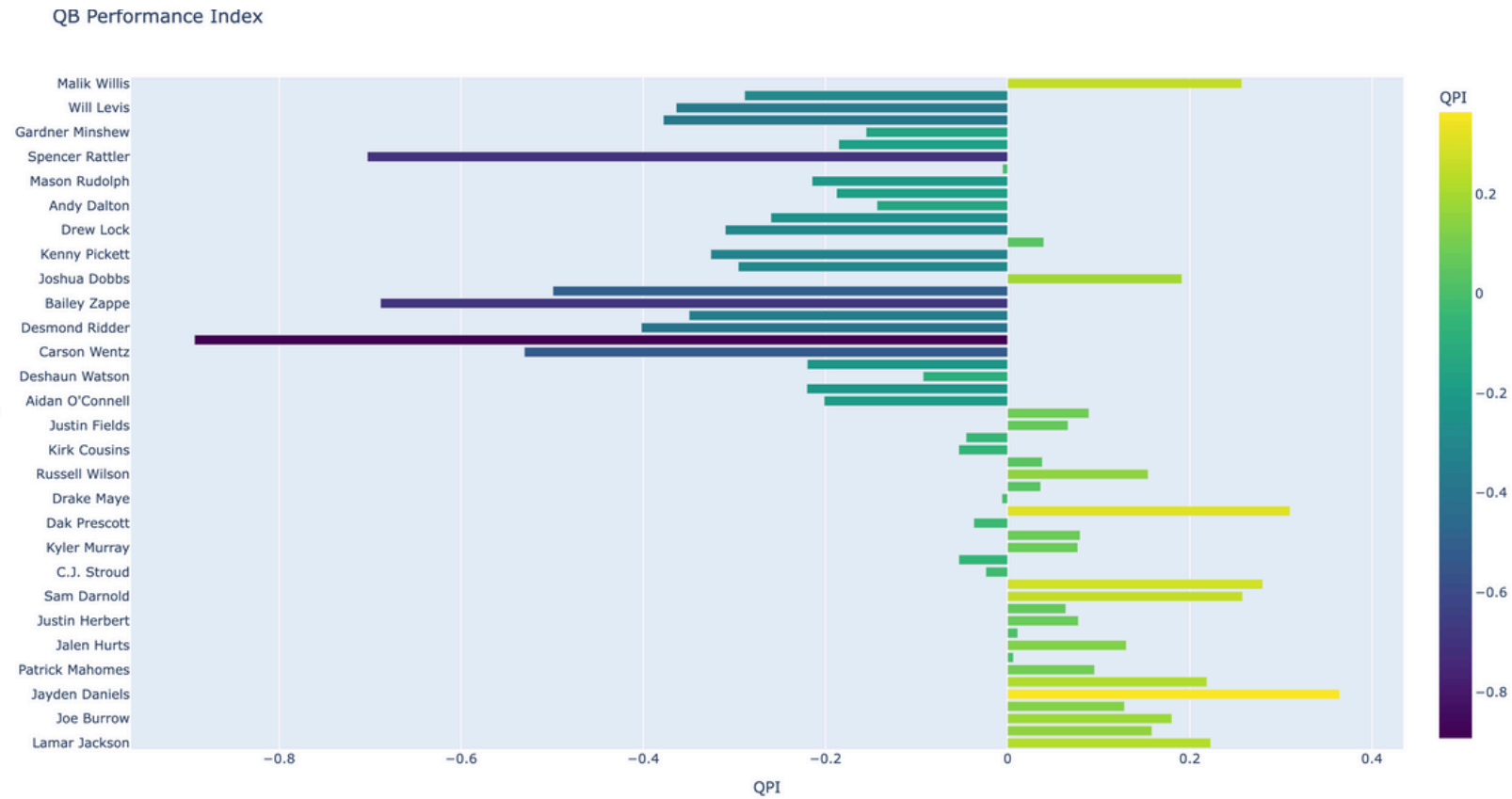
Quarterback
Performance Index:

Our QPI combines individual statistics, clutch ability (redzone conversion rate), and salary to create a cumulative metric

Top 5 Undervalued QBs (High QPI per salary):		
Player	Salary	QPI
Jayden Daniels	9436663	0.3628
Bo Nix	4653292	0.2979
Brock Purdy	934252	0.2791
Malik Willis	1290025	0.2561
Sam Darnold	10000000	0.2546
Lamar Jackson	52000000	0.2279
Baker Mayfield	33333333	0.2197
Joshua Dobbs	2250000	0.1901
Joe Burrow	55000000	0.1802
Josh Allen	43005667	0.1606

Top 5 Overvalued QBs (Low QPI per salary):		
Player	Salary	QPI
Jake Haener	1136204	-0.8933
Spencer Rattler	1089120	-0.7032
Bailey Zappe	985000	-0.6892
Carson Wentz	3325000	-0.5306
Brandon Allen	2020000	-0.5000
Desmond Ridder	985000	-0.4027
Tyler Huntley	1125000	-0.3783
Will Levis	2385541	-0.3644
Trey Lance	8526319	-0.3499
Kenny Pickett	3516976	-0.3264

$$\frac{(0.15 * pyd + 0.15 * comp - 0.20 * int + 0.05 * rtd + 0.05 * ryd + 0.15 * clutchfactor)}{(1 + 0.0000001 * salary)}$$



Cluster Summary Statistics:		
Cluster	Passing Yards	Rushing Yards
0	2653.90	146.25
1	685.00	60.18
2	4036.77	439.85

Cluster	Completion Percentage \	Passing Touchdowns	Interceptions	Clutch Score	Salary	Num_QBs
0	64.63	16.00	8.80	28.19	25227608.20	20
1	61.92	3.32	2.18	26.60	4967841.68	22
2	68.27	29.62	10.31	33.61	32958708.23	13

Coefficients: [0.78644949 0.33432514 0.03238241 0.18624272 -0.19353507]

We used a Linear Regression model to analyze how much of an effect different metrics have on salary

Coefficients: [Passing Yards, Passing Touchdowns, Interceptions, Completion Percentage, Clutch Factor]

Passing Yards have the highest positive correlation, and clutch factor surprisingly has a negative one!

Objectives

1. Help GMs identify their target QBs
2. Understand what makes a QB higher or lower valued
3. Identify undervalued and overvalues QBs
4. Build a cumulative QPI that holistically ranks QBs

K Means - This chart shows three QB archetypes:

pink = backups/low-volume players,
blue = efficient pocket passers
yellow = elite franchise QBs with strong passing and rushing.

These clusters help teams benchmark compensation and target QBs that fit their offensive style. Clusters help benchmark QB value showing who's overpaid or undervalued and what teams pay for: efficiency (C0), depth (C1), or dual-threat impact (C2)